



**SILVER OAK
UNIVERSITY**
EDUCATION TO INNOVATION



**SILVER OAK
UNIVERSITY**
EDUCATION TO INNOVATION

Silver Oak College of Engineering and Technology
B.Tech CE Minor Project / Advance Project – Weekly Review

Section A: Details of Student		
1.	Name(s) of Student(s): (For multiple students, enter names separated by commas.)	MIHIR KASUNDRA, SOLANKI GAURAV
2.	Enrollment No.: (For multiple students, enter Enrollment No separated by commas.)	2402030430084, 2402030430110
3.	Department	Computer Engineering
4.	Group/Individual	Group (Project ID: M2_CE_045)
5.	Minor Project / Advance Project Phase: Phase I / Phase II	Minor Project - Phase I
6.	Project Title	LinguaAI: Speech-to-Text & AI-Powered Endangered Language Preservation Platform
7.	Name of the Guide	Ms. JANKI BAROT
8.	Name of the Co-Guide (if any)	N/A
9.	Week No. & Dates: (e.g., Week 5: 1–7 Jan 2026)	Week 6: 20–26 Sep 2026
10.	Date(s) of previous meetings of Guide	19/09/2026

Section B: Objectives Planned for the Week		
9.	Previous MINOR PROJECT / ADVANCE PROJECT Review date and progress (milestone) achieved as on that date	Date: 19/09/2026 Milestone Achieved: Finalized project scope, completed initial literature search on Web Speech API, designed database schema, and prepared presentation slides.
10.	What was planned at the start of the week	1. Integrate JavaScript Web Speech API for live microphone transcription. 2. Set up local MySQL database schema ('users', 'languages', 'transcriptions'). 3. Implement user signup/login module in PHP with session handling.

Work Completed During the Week		
11.	Actual progress Made	1. Successfully integrated browser SpeechRecognition API in 'index.php' with real-time text output. 2. Created MySQL database tables with utf8mb4_unicode_ci collation for multilingual scripts. 3. Built user login/signup system with password hashing. 4. Connected frontend Speech interface to PHP backend via AJAX Fetch API.

Literature Review Progress		
12.	New Papers Studied this Week: (YES/NO)	YES
13.	Total papers reviewed till date: (Target: minimum 4–5 core papers)	5 core papers
14.	Key learning / gap identified	Key Learning: Web Speech API processes speech client-side without heavy GPU server costs. Gap Identified: Existing tools lack integrated database tracking for endangered language status badges (Safe, Vulnerable, Endangered).

Scope & Problem Definition Status		
15.	• Problem statement finalized • Scope defined • Objectives refined • Changes suggested this week (if any):	• Problem statement finalized: Yes • Scope defined: Yes (Focusing on real-time web speech-to-text & language cataloging) • Objectives refined: Yes • Changes suggested: Enforce strict UTF-8 table collation in MySQL to prevent character corruption in non-Latin scripts.

Section C: Implementation / Experimentation Status		
16.	Module / Algorithm / Model implemented this week:	• Client-Side Speech Transcriber Module ('index.php' + JS Speech API) • DB Auth Handler ('config.php', 'login.php') • AJAX Save Endpoint ('backend.php')
17.	Tools / Technologies used:	HTML5, CSS3, JavaScript (ES6+), W3C Web Speech API, PHP 8.x, MySQL/MariaDB, XAMPP Server, VS Code.
18.	Results obtained (if any):	Achieved real-time speech-to-text conversion for English and Hindi locales in browser with ~100ms latency. Transcriptions successfully saved to local MySQL table.
19.	Pending implementation work:	• Admin moderation panel ('admin.php') • Visual analytics charts using Chart.js ('dashboard.php') • UTF-8 BOM CSV data exporter script • Cross-browser compatibility testing

Industry / Expert Interaction (if applicable)		
20.	Visit / Meeting conducted this week: Yes / No	No
21.	Expert / Industry Name:	N/A
22.	Mode: Visit / Online / Email	N/A
23.	Purpose & outcome:	N/A

Issues / Challenges Faced		
24.		1. Database character encoding issue: Regional language characters showed as '?' until MySQL table collation was changed to utf8mb4_unicode_ci. 2. Browser microphone permission prompts blocking continuous speech recognition in Chrome.

Section D: Guide's Suggestions & Corrections (To be filled during review)		
25.		

Plan for Next Week (Clear, achievable tasks) (To be filled during review)		
26.		

Overall Progress Status		
27.	Completion Percentage (approx.): %	
28.	Progress Rating: Excellent / Very Good / Good / Satisfactory / Needs Improvement	

Name of the Students	Signature	Date
MIHIR KASUNDRA, SOLANKI GAURAV		26/09/2026
Name of the Guide	Signature	Date
Ms. JANKI BAROT		26/09/2026